HILBERT'S INEQUALITY

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1. *Introduction*

Hilbert demonstrated that

$$\left| \sum_{r \neq s} \frac{u_r \bar{u}_s}{r-s} \right| \leqslant 2\pi \Sigma |u_r|^2 \tag{1.1}$$

for any set of complex u_r, and later Schur [13] replaced the 2π by the best possible constant π. In this paper we obtain a precise bound for the more general bilinear forms

$$\sum_{r \neq s} u_r \bar{u}_s \csc \pi(x_r - x_s), \quad \sum_{r \neq s} u_r \bar{u}_s / (\lambda_r - \lambda_s).$$

These bilinear forms arise in various questions of function theory and analytic number theory. Here $1 \leqslant r \leqslant R$, $1 \leqslant s \leqslant R$, the x_r are real and distinct modulo 1, the λ_r are real and distinct, and $\csc \theta = 1/\sin \theta$. These forms have been considered previously by Titchmarsh [14], who applied his bounds in harmonic analysis, and more recently by Elliott [2], Kobayashi [7] and Matthews [9] in connection with the large sieve.

Before proceeding further we simplify matters by introducing some notation. When f ranges over a finite set of non-negative values, $\min_+ f$ denotes the least positive value. Summations are from 1 to R unless the contrary is indicated, and the symbol $\sum'$ denotes a sum in which the variables are restricted to exclude infinite summands. As usual, $\|\theta\|$ denotes the distance from θ to the nearest integer, that is, $\|\theta\| = \min_n |\theta - n|$.

THEOREM 1. *Let $x_1, x_2, \ldots, x_R$ denote real numbers which are distinct modulo 1, and suppose that*

$$\delta = \min_{r,s}{}_+ \|x_r - x_s\|.$$

Then

$$\left| \sum_{r,s}{}' u_r \bar{u}_s \csc \pi(x_r - x_s) \right| \leqslant \delta^{-1} \sum_r |u_r|^2. \tag{1.2}$$

Moreover, if

$$0 < \delta_r \leqslant \min_s{}_+ \|x_r - x_s\|,$$

then

$$\left| \sum_{r,s}{}' u_r \bar{u}_s \csc \pi(x_r - x_s) \right| \leqslant \tfrac{1}{2} \sum_r |u_r|^2 \delta_r^{-1}. \tag{1.3}$$

It follows from a paper of Hellinger and Toeplitz [5; §2] (see also §8.9 of [4]) that (1.3) holds also for infinite sums, provided that $\min_+ f$ is replaced by $\inf_+ f$. This remark holds equally for (1.5), (1.6) and (1.7) below. On the other hand, it

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is easily seen that $\delta = 0$ for an infinite set of x_r, distinct modulo 1, so that (1.2) and (1.4) become vacuous.

The factor δ^{-1} in (1.2) can be replaced by $\delta^{-1} - C$ for some positive C, but from the work of Bombieri and Davenport [1] (or from an unpublished theorem of Matthews) it follows that it is not possible to replace δ^{-1} by $\delta^{-1} - C$ if $C > 1$. In (1.3) the constant is not best possible. It may be that (1.3) holds with $\frac{3}{2}$ replaced by unity, and then it would obviously contain (1.2).

In a separate paper [11], we use (1.2) to obtain a definitive formulation of the large sieve, and from (1.3) we derive a delicate sieve which provides improved estimates in some situations.

Corollary 1 is the analogue of Theorem 1 with cotangent replacing cosecant. It is possible to prove this corollary directly and take it as fundamental, deriving Theorem 1 from it.

COROLLARY 1. *On the hypotheses of Theorem* 1,

$$\left| \sum_{r,s}{}' u_r \bar{u}_s \cot \pi(x_r - x_s) \right| \leqslant \delta^{-1} \sum_r |u_r|^2 \tag{1.4}$$

and

$$\left| \sum_{r,s}{}' u_r \bar{u}_s \cot \pi (x_r - x_s) \right| \leqslant \tfrac{3}{2} \sum_r |u_r|^2 \delta_r^{-1}. \tag{1.5}$$

Corresponding to Theorem 1 there is a result for the real line, which may be proved in a similar manner. To avoid being repetitious, we derive Theorem 2 from Theorem 1 by a simple limiting argument.

THEOREM 2. *Suppose that* $\lambda_1, \lambda_2, \ldots, \lambda_R$ *are distinct real numbers, and*

$$\delta = \min_{r,s}{}_+ |\lambda_r - \lambda_s|.$$

Then

$$\left| \sum_{r,s}{}' u_r \bar{u}_s/(\lambda_r - \lambda_s) \right| \leqslant \pi \delta^{-1} \sum_r |u_r|^2. \tag{1.6}$$

Furthermore, if

$$\delta_r = \min_s{}_+ |\lambda_r - \lambda_s|,$$

then

$$\left| \sum_{r,s}{}' u_r \bar{u}_s/(\lambda_r - \lambda_s) \right| \leqslant \tfrac{3}{2}\pi \sum_r |u_r|^2 \delta_r^{-1}. \tag{1.7}$$

One consequence of this concerns almost periodic polynomials.

COROLLARY 2. *On the premises of Theorem* 2,

$$\int_0^T \left| \sum_r a_r \exp(i\lambda_r t) \right|^2 dt = (T + 2\pi\theta_1 \delta^{-1}) \sum_r |a_r|^2 \tag{1.8}$$

and

$$\int_0^T \left| \sum_r a_r \exp(i\lambda_r t) \right|^2 dt = \sum_r |a_r|^2 (T + 3\pi\theta_2 \delta_r^{-1}). \tag{1.9}$$

Here the θ_n depend on the various parameters in such a manner that $|\theta_n| \leqslant 1$.

We note that (1.8) and (1.9) also hold for infinite sums whenever the respective right-hand sides are finite, for then the case of finite sums implies that the partial sums on the left converge in mean square to a function in L_2 (see Titchmarsh [**15**; p. 387]).

If we suppose that $T \geqslant 2\pi(1+\varepsilon)\delta^{-1}$, then it is immediate from (1.8) that

$$\frac{1}{T}\int_0^T \left|\sum_r a_r \exp(i\lambda_r t)\right|^2 dt \geqslant C(\varepsilon)\sum_r |a_r|^2. \tag{1.10}$$

With the stronger condition $T \geqslant 9\pi(1+\varepsilon)\delta^{-1}$, this was first established by Wiener [**17**] (see also Paley and Wiener [**12**; p. 125]). From (1.10) Wiener easily deduced his gap theorem for power series, namely that if the μ_n are natural numbers such that $\mu_{n+1} - \mu_n \to \infty$ as $n \to \infty$, and if the power series

$$f(z) = \sum_n a_n z^{\mu_n}$$

has radius of convergence 1, then the circle $|z| = 1$ is a natural boundary of f. Marcinkievic and Zygmund [**8**] have given a short proof of (1.10) when $T \geqslant 4\pi(1+\varepsilon)\delta^{-1}$. Ingham [**6**] weakened the condition to $T \geqslant 2\pi(1+\varepsilon)\delta^{-1}$, and showed that $C(\varepsilon)$ must tend to 0 with ε. Nevertheless, it seems to have remained unobserved that (1.10) could be derived from a result of the kind (1.8). A weaker form of (1.8) has been given by Titchmarsh [**14**] who gave other applications to harmonic analysis.

The estimate (1.9) is of particular interest when applied to ordinary Dirichlet series.

COROLLARY 3. *If* $\sum_{n=1}^{\infty} n|a_n|^2 < \infty$, *then*

$$\int_0^T \left|\sum_{n=1}^{\infty} a_n n^{-it}\right|^2 dt = \sum_{n=1}^{\infty} |a_n|^2 (T + O(n)), \tag{1.11}$$

the implied constant being absolute.

This sharpens earlier mean-value theorems for Dirichlet series, since previously (see [**10**; p. 50]) one knew only that

$$\int_0^T \left|\sum_{n=1}^{N} a_n n^{-it}\right|^2 dt = (T + O(N))\sum_{n=1}^{N} |a_n|^2. \tag{1.12}$$

The relative power of (1.11) may be demonstrated by the observation that the simple formula (see [**16**; Theorem 4.11])

$$\zeta(\tfrac{1}{2}+it) = \sum_{n \leqslant N} n^{-\frac{1}{2}-it} + O(N^{\frac{1}{2}}/(1+|t|)), \tag{1.13}$$

which holds for $|t| \leqslant N$, combines with (1.11) to establish

$$\int_0^T |\zeta(\tfrac{1}{2}+it)|^2 dt = T \log T + O(T). \tag{1.14}$$

Of course, (1.14) is classical (see Theorem 2.4.1 of [3]), but other proofs (as found in Chapter 7 of [15]) require a deeper knowledge of the behaviour of the zeta function. In fact, since (1.14) does not follow in the same way from (1.12), Titchmarsh was led to assert [16; p. 118] that (1.13) does not suffice to give (1.14).

We are happy to record our gratitude to Professor Atle Selberg, who communicated to us several important ideas which we use below.

2. *Lemmata*

We establish here a number of simple inequalities which we use in the proof of Theorem 1. Only the first lemma appears in the proof of (1.2); all the others pertain to (1.3).

LEMMA 1. *The inequality*

$$\csc^2 \pi x + 2|\cot \pi x \csc \pi x| \leqslant 3\pi^{-2}\|x\|^{-2}$$

holds for all real x.

Proof. It suffices to show that

$$3\sin^2\theta - \theta^2(1+2\cos\theta) \geqslant 0$$

when $0 \leqslant \theta \leqslant \frac{1}{2}\pi$. It is easy to see that the left-hand side may be written in the form

$$\sum_{k=3}^{\infty} \frac{a_k \theta^{2k}}{(2k)!},$$

where

$$a_k = (-1)^{k+1}(3.2^{2k-1} - 4k(2k-1)).$$

The terms alternate in sign and are strictly decreasing in magnitude for $|\theta| \leqslant 2$, so that the sum is positive for $|\theta| \leqslant 2$. This is more than we require.

LEMMA 2. *On the hypotheses of Theorem* 1,

$$\sum_r{}' \delta_r \csc^2 \pi(x_r - x_s) < 4\pi^{-2}\delta_s^{-1}.$$

Proof. Since $\csc^2 \pi x$ is a convex function of x for $0 < x < 1$, each term of the sum satisfies

$$\delta_r \csc^2 \pi(x_r - x_s) \leqslant \int_{x_r - \frac{1}{2}\delta_r}^{x_r + \frac{1}{2}\delta_r} \csc^2 \pi(x - x_s)\, dx.$$

Hence the sum in question is at most

$$\int_{\frac{1}{2}\delta_s}^{1-\frac{1}{2}\delta_s} \csc^2 \pi x dx = 2\pi^{-1}\cot \tfrac{1}{2}\pi\delta_s < 4\pi^{-2}\delta_s^{-1}.$$

LEMMA 3. *Let*

$$f(\delta) = \delta \csc^4 \pi\delta + \pi^{-1}\cot \pi\delta + \tfrac{1}{3}\pi^{-1}\cot^3 \pi\delta.$$

If $0 < \delta \leqslant \frac{1}{6}$, *then*

$$f(\delta) \leqslant \tfrac{7}{4}\pi^{-4}\delta^{-3}. \tag{2.1}$$

Proof. It suffices to show that

$$f(\delta) \leqslant \tfrac{4}{3}\delta((\pi\delta)^{-4} + (\pi\delta)^{-2}) + 4\delta^3 \tag{2.2}$$

whenever $0 < \delta \leqslant \frac{1}{2}$. For then we only have to note the inequality $\pi^2 < 10$ to see that

$$f(\delta) < \pi^{-4}\delta^{-3}\left(\tfrac{4}{3}+\tfrac{10}{27}+\tfrac{1}{36}\right)$$

when $0 < \delta \leqslant \frac{1}{6}$, and this gives (2.1).

We show (2.2) be recalling the partial fraction formulae below, in the first of which the terms n and $-n$ are grouped together;

$$\cot z = \sum_{n=-\infty}^{\infty} (z-\pi n)^{-1},$$

$$\sin^{-2} z = \sum_{n=-\infty}^{\infty} (z-\pi n)^{-2},$$

$$\cot^3 z + \cot z = \sum_{n=-\infty}^{\infty} (z-\pi n)^{-3},$$

$$\sin^{-4} z - \tfrac{2}{3}\sin^{-2} z = \sum_{n=-\infty}^{\infty} (z-\pi n)^{-4}.$$

We apply these formulae to f, write the term $n = 0$ separately and collect the terms n and $-n$ as one. This gives

$$f(\delta) - \tfrac{4}{3}\delta\left((\pi\delta)^{-4} + (\pi\delta)^{-2}\right) = \delta^3 \sum_{n=1}^{\infty} g(n,\delta) \tag{2.3}$$

where $g(n,\delta)$ is a sum of three expressions of the form $c\delta^a(n^2-\delta^2)^{-b}$ with non-negative a, b, c. Since $\delta^a\ (n^2-\delta^2)^{-b}$ are increasing for $0 < \delta \leqslant \frac{1}{2}$, the sum on the right of (2.3) is an increasing function of δ. Its value at $\delta = \frac{1}{2}$ is easily computed by examining the left-hand side, and this establishes (2.2) and completes the proof of the lemma.

LEMMA 4. *If* $0 < \delta_r \leqslant \frac{1}{6}$, *then*

$$\sum_r{}' \delta_r \csc^4 \pi(x_r - x_s) < \tfrac{7}{2}\pi^{-4}\delta_s^{-3}.$$

Proof. We assume as we may that the x_r are in increasing order in $[0, 1)$. Then we put $x_{R+1} = x_1$ and $\delta_{r+\frac{1}{2}} = \max(\delta_r, \delta_{r+1})$. The function $\csc^4 \pi x$ is convex, so that the sum to be estimated is no larger than

$$\delta_{s+\frac{1}{2}} \csc^4 \pi\delta_{s+\frac{1}{2}} + \delta_{s-\frac{1}{2}} \csc^4 \pi\delta_{s-\frac{1}{2}} + \int_{\delta_{s+1/2}}^{1-\delta_{s-1/2}} \csc^4 \pi x\,dx.$$

Since $\csc^4 \pi x = -\pi^{-1}(\cot \pi x + \frac{1}{3}\cot^3 \pi x)'$, the above is equal to $f(\delta_{s+\frac{1}{2}}) + f(\delta_{s-\frac{1}{2}})$ in the notation of Lemma 3. Lemma 4 is now a consequence of (2.1).

LEMMA 5. *Suppose that* $s \neq t$. *Then*

$$\sum_r{}' \delta_r \csc^2 \pi(x_r - x_s) \csc^2 \pi(x_r - x_t) \leqslant 4\pi^{-2}(\delta_s^{-1} + \delta_t^{-1}) \csc^2 \pi(x_s - x_t).$$

Proof. We may assume that $c < x_1 < x_2 < \ldots < x_R < 1 + c$, and that $c \leqslant x_s - \frac{1}{2}\delta_s < x_s < x_t < x_t + \frac{1}{2}\delta_t \leqslant 1 + c$. Let $\mathscr{A}$ denote the set of those points x for

which $c \leqslant x \leqslant 1+c$, $|x-x_s| \geqslant \frac{1}{2}\delta_s$ and $|x-x_t| \geqslant \frac{1}{2}\delta_t$. As $\csc^2 \pi(x-x_s) \csc^2 \pi(x-x_t)$ is convex, the sum under consideration is majorized by

$$\int_{\mathscr{A}} \csc^2 \pi(x-x_s) \csc^2 \pi(x-x_t)\, dx.$$

Let $\mathscr{B}_r$ be the semicircular path of radius $\frac{1}{2}\delta_r$ and centre x_r passing from $x_r+\frac{1}{2}\delta_r$ through $x_r+\frac{1}{2}i\delta_r$ to $x_r-\frac{1}{2}\delta_r$. It is easily shown that $|\csc(x+iy)| \leqslant |y|^{-1}$, so that

$$\int_{\mathscr{A}} = \int_{\mathscr{B}_s} + \int_{\mathscr{B}_t} + \int_{c}^{c+i\infty} + \int_{1+c+i\infty}^{1+c}.$$

Moreover, $\csc^2 \pi z$ has period 1, so that the last two integrals cancel. Now consider the integral along $\mathscr{B}_s$. Write the integrand as

$$\pi^{-2}(z-x_s)^{-2} \csc^2 \pi(z-x_t) + \left(\csc^2 \pi(z-x_s) - \pi^{-2}(z-x_s)^{-2}\right) \csc^2 \pi(z-x_t)$$

which gives rise to two integrals, say

$$\int_{\mathscr{B}_s} = I + J.$$

The integrand of J has a removable singularity at x_s, so that the semicircular path may be replaced by the line segment from $x_s+\frac{1}{2}\delta_s$ to $x_s-\frac{1}{2}\delta_s$. Also, the integrand of J is non-negative on the real axis. Hence $J \leqslant 0$. Hence

$$\int_{\mathscr{A}} = \int_{\mathscr{B}_s} + \int_{\mathscr{B}_t} \leqslant I(\tfrac{1}{2}\delta_s, x_s-x_t) + I(\tfrac{1}{2}\delta_t, x_t-x_s), \tag{2.4}$$

where

$$I(\rho, \alpha) = i\rho^{-1} \pi^{-2} \int_0^{\pi} \left(\csc \pi(\alpha+\rho e^{i\theta})\right)^2 e^{-i\theta}\, d\theta. \tag{2.5}$$

To estimate $I(\rho, \alpha)$, we write

$$\csc^2 \pi(\alpha+z) = \sum_{k=0}^{\infty} c_k z^k.$$

This expansion is valid in the disc $|z| < \|\alpha\|$ and it converges uniformly for $|z| \leqslant \frac{1}{2}\|\alpha\|$. The coefficients $c_k = c_k(\alpha)$ are real, $c_0 = \csc^2 \pi\alpha$ and $c_{2k} \geqslant 0$ for any real α. This last assertion may be proved by induction, but it is immediate from the formula

$$\frac{d^k}{dz^k} \csc^2 \pi z = (-1)^k \pi^{-2}(k+1)! \sum_{n=-\infty}^{\infty} (z-n)^{-2-k}.$$

Thus the integral in (2.5) may be written

$$\sum_{k=0}^{\infty} c_k \rho^k \int_0^{\pi} \exp\left(i(k-1)\theta\right) d\theta = \pi c_1 \rho + 2i \sum_{k=0}^{\infty} \frac{c_{2k} \rho^{2k}}{2k-1},$$

whence

$$\operatorname{Re} I(\rho, \alpha) = \frac{2c_0}{\pi^2 \rho} - \frac{2}{\pi^2} \sum_{k=1}^{\infty} \frac{c_{2k} \rho^{2k-1}}{2k-1} \leqslant \frac{2}{\pi^2 \rho} \csc^2 \pi\alpha.$$

This with (2.4) gives the required inequality.

3. *Proof of Theorem* 1

Our proof is modelled on Schur's proof [**13**] of Hilbert's inequality. We first prove (1.2). The bilinear form under consideration is skew-Hermitian. Thus there is an extremal set of the u_r for which

$$\sum_r{}' u_r \csc \pi(x_r - x_s) = i\mu u_s \tag{3.1}$$

for $1 \leqslant s \leqslant R$. Also, we may normalize so that

$$\sum_r |u_r|^2 = 1. \tag{3.2}$$

It suffices to bound the form with values of u_r satisfying (3.1) and (3.2).

By Cauchy's inequality,

$$\begin{aligned}\left|\sum_{r,s}{}' u_r \bar{u}_s \csc \pi(x_r - x_s)\right|^2 &\leqslant \sum_r \left|\sum_s{}' \bar{u}_s \csc \pi(x_r - x_s)\right|^2 \\ &= \sum_{s,t} \bar{u}_s u_t \sum_r{}' \csc \pi(x_r - x_s) \csc \pi(x_r - x_t) \\ &= S_1 + S_2, \end{aligned} \tag{3.3}$$

where

$$S_1 = \sum_s |u_s|^2 \sum_r{}' \csc^2 \pi(x_r - x_s) \tag{3.4}$$

and

$$S_2 = \sum_{s \neq t} \bar{u}_s u_t \sum_r{}' \csc \pi(x_r - x_s) \csc \pi(x_r - x_t). \tag{3.5}$$

In S_2 the variables r, s and t are distinct, so that we may write

$$\csc \pi(x_r - x_s) \csc \pi(x_r - x_t) = \csc \pi(x_s - x_t)(\cot \pi(x_r - x_s) - \cot \pi(x_r - x_t)). \tag{3.6}$$

We use this to split S_2 into two parts. In the first of these a term with $r = t$ can be added and in the second a term with $r = s$. Hence $S_2 = S_3 - S_4 + 2 \operatorname{Re} S_5$, where

$$S_3 = \sum_{r,s,t}{}' \bar{u}_s u_t \csc \pi(x_s - x_t) \cot \pi(x_r - x_s), \tag{3.7}$$

$$S_4 = \sum_{r,s,t}{}' \bar{u}_s u_t \csc \pi(x_s - x_t) \cot \pi(x_r - x_t) \tag{3.8}$$

and

$$S_5 = \sum_{r,s}{}' \bar{u}_s u_r \csc \pi(x_r - x_s) \cot \pi(x_r - x_s). \tag{3.9}$$

We show now that $S_3 = S_4$. We see from (3.7) and (3.1) that

$$S_3 = \sum_{r,s}{}' \bar{u}_s \cot \pi(x_r - x_s)(-i\mu u_s) = -i\mu \sum_{r,s}{}' |u_s|^2 \cot \pi(x_r - x_s).$$

Similarly, from (3.8) and (3.1),

$$S_4 = \sum_{r,t}{}' u_t \cot \pi(x_r - x_t)(-i\mu \bar{u}_t) = -i\mu \sum_{r,t}{}' |u_t|^2 \cot \pi(x_r - x_t).$$

Therefore $S_3 = S_4$, so that $S_1 + S_2 = S_1 + 2 \operatorname{Re} S_5 \leqslant S_1 + 2|S_5|$. We use the inequality $2|u_s u_r| \leqslant |u_s|^2 + |u_r|^2$ in (3.9), so that (3.4) and (3.9) give

$$S_1 + S_2 \leqslant \sum_{r,s}{}' |u_r|^2 (\csc^2 \pi(x_r - x_s) + 2|\csc \pi(x_r - x_s) \cot \pi(x_r - x_s)|),$$

and by Lemma 1 this is

$$\leqslant 3\pi^{-2} \sum_r |u_r|^2 \sum_s{}' \|x_r - x_s\|^{-2}.$$

The x_r are spaced from each other by at least δ, so that

$$\sum_s{}' \|x_r - x_s\|^{-2} < 2 \sum_{k=1}^{\infty} (k\delta)^{-2} = \tfrac{1}{3}\pi^2 \delta^{-2}.$$

We combine this with (3.2) to obtain $S_1 + S_2 < \delta^{-2}$, and then an appeal to (3.3) completes the proof of (1.2). A careful examination of the above argument reveals that it suffices to consider k in the range $1 \leqslant k \leqslant \frac{1}{2}(R+1)$. This leads to a sharper upper bound of the form $\delta^{-1} - C$.

We now prove (1.3). We first establish (1.3) when all the δ_r satisfy $\delta_r \leqslant \frac{1}{6}$. Our argument is on the same lines as the one above. This time we extremalize and normalize so that

$$\sum_r{}' u_r \csc \pi(x_r - x_s) = i\mu u_s \delta_s^{-1} \tag{3.10}$$

for $1 \leqslant s \leqslant R$ and

$$\sum_r |u_r|^2 \delta_r^{-1} = 1. \tag{3.11}$$

Cauchy's inequality now gives

$$\left| \sum_{r,s}{}' u_r \bar{u}_s \csc \pi(x_r - x_s) \right|^2 \leqslant T_1 + T_2, \tag{3.12}$$

where

$$T_1 = \sum_{r,s}{}' \delta_r |u_s|^2 \csc^2 \pi(x_r - x_s) \tag{3.13}$$

and

$$T_2 = \sum_{s \neq t} \bar{u}_s u_t \sum_r{}' \delta_r \csc \pi(x_r - x_s) \csc \pi(x_r - x_t). \tag{3.14}$$

As before, we apply the identity (3.6) and obtain $T_2 = T_3 - T_4 + 2 \operatorname{Re} T_5$, where

$$T_3 = \sum_{s,t,r}{}' \delta_r \bar{u}_s u_t \csc \pi(x_s - x_t) \cot \pi(x_r - x_s), \tag{3.15}$$

$$T_4 = \sum_{s,t,r}{}' \delta_r \bar{u}_s u_t \csc \pi(x_s - x_t) \cot \pi(x_r - x_t) \tag{3.16}$$

and

$$T_5 = \sum_{r,s}{}' \delta_r \bar{u}_s u_r \csc \pi(x_r - x_s) \cot \pi(x_r - x_s). \tag{3.17}$$

When the sums T_3 and T_4 are simplified by reference to (3.10), we find once more that $T_3 = T_4$. Hence $T_2 = 2 \operatorname{Re} T_5$. We replace the cotangent in (3.17) by cosecant so that $T_2 \leqslant T_6$, where

$$T_6 = \sum_{r,s}{}' |u_r u_s| (\delta_r + \delta_s) \csc^2 \pi(x_r - x_s).$$

We apply Cauchy's inequality to T_6, and use (3.11) to obtain

$$(\tfrac{1}{2}T_6)^2 \leqslant \sum_r \delta_r \left(\sum_s{}' |u_s| \delta_s \csc^2 \pi(x_r - x_s) \right)^2$$

$$= \sum_{s,t} |u_s u_t| \delta_s \delta_t \sum_r{}' \delta_r \csc^2 \pi(x_r - x_s) \csc^2 \pi(x_r - x_t).$$

We split this last expression into two parts, Σ_1 and Σ_2, where Σ_1 contains the terms with $s = t$ and Σ_2 the balance. It is immediate from Lemma 4 and (3.11) that

$\sum_1 \leqslant \frac{7}{2}\pi^{-4}$, and Lemma 5 gives $\sum_2 \leqslant 4\pi^{-2}T_6$. Taken together these inequalities give the bound

$$T_6^2 \leqslant 14\pi^{-4}+16\pi^{-2}\,T_6,$$

so that

$$(T_6-8\pi^{-2})^2 \leqslant 78\pi^{-4} < (9\pi^{-2})^2.$$

Thus T_2 satisfies the inequality $T_2 \leqslant T_6 \leqslant 17\pi^{-2}$. We majorize the sum T_1, given by (3.13), by an appeal to Lemma 2 which yields $T_1 \leqslant 4\pi^{-2}$. Hence $T_1+T_2 \leqslant 21\pi^{-2}$ and this with (3.12) enables us to assert (1.3) when all the δ_r satisfy $\delta_r \leqslant \frac{1}{6}$.

To prove (1.3) when some of the δ_r satisfy $\delta_r > \frac{1}{6}$ we argue as follows. Let $n = 3$ or 2. We suppose that (1.3) is true whenever $\delta_r \leqslant 2^{-n}$ and show that it is true whenever $\delta_r \leqslant 2^{1-n}$. We first note that

$$2\csc\pi\theta = \cot\tfrac{1}{2}\pi\theta-\cot\tfrac{1}{2}\pi(\theta+1),$$

$$4\csc\pi\theta = \left(\exp(\tfrac{1}{2}\pi i\theta)+\exp(-\tfrac{1}{2}\pi i\theta)\right)\csc\tfrac{1}{2}\pi\theta - \left(\exp\left(\tfrac{1}{2}\pi i(\theta+1)\right)+\exp\left(-\tfrac{1}{2}\pi i(\theta+1)\right)\right)\csc\tfrac{1}{2}\pi(\theta+1)$$

and

$$\begin{aligned}8\csc\pi\theta = {} & 2\exp(\tfrac{1}{2}\pi i\theta)\csc\tfrac{1}{2}\pi\theta-\exp\left(\tfrac{1}{2}\pi i(\theta+1)\right)\csc\tfrac{1}{2}\pi(\theta+1)\\ & -\exp\left(\tfrac{1}{2}\pi i(\theta-1)\right)\csc\tfrac{1}{2}\pi(\theta-1)+2\exp(-\tfrac{1}{2}\pi i\theta)\csc\tfrac{1}{2}\pi\theta\\ & -\exp\left(-\tfrac{1}{2}\pi i(\theta+1)\right)\csc\tfrac{1}{2}\pi(\theta+1)-\exp\left(-\tfrac{1}{2}\pi i(\theta-1)\right)\csc\tfrac{1}{2}\pi(\theta-1).\end{aligned}$$

Therefore

$$8\sum_{r,s}{}' u_r\bar{u}_s\csc\pi(x_r-x_s) = A_1+A_2, \tag{3.18}$$

where

$$A_m = \sum_{r=1}^{2R}\sum_{s=1}^{2R} u_r^{(m)}\bar{u}_s^{(m)}\csc\pi(x_r^{(m)}-x_s^{(m)}) \tag{3.19}$$

with

$$u_{2r}^{(m)} = u_r\exp\left(\tfrac{1}{2}\pi i(-1)^{m-1}x_r\right),\quad u_{2r-1}^{(m)} = -u_r\exp\left(\tfrac{1}{2}\pi i(-1)^{m-1}(1+x_r)\right),$$

$$x_{2r}^{(m)} = \tfrac{1}{2}x_r \text{ and } x_{2r-1}^{(m)} = \tfrac{1}{2}(1+x_r).$$

We define

$$\delta_{2r}^{(m)} = \delta_{2r-1}^{(m)} = \tfrac{1}{2}\delta_r,$$

so that if $\delta_r \leqslant 2^{1-n}$ and $0 < \delta_r \leqslant \min_s{}_+ \|x_r-x_s\|$, then $\delta_r^{(m)} \leqslant 2^{-n}$ and

$$0 < \delta_r^{(m)} \leqslant \min_s{}_+ \|x_r^{(m)}-x_s^{(m)}\|.$$

Hence, by (3.19) and our assumption that (1.3) holds whenever $\delta_r \leqslant 2^{-n}$, we have

$$|A_m| \leqslant \sum_{r=1}^{2R}|u_r^{(m)}|^2/\delta_r^{(m)} = 4\sum_r |u_r|^2\delta_r{}^{-1}.$$

This with (3.18) establishes (1.3) whenever $\delta_r \leqslant 2^{1-n}$.

4. *Theorem 2 and the Corollaries*

The bilinear form in Corollary 1 can be written as a linear combination of bilinear forms of the type considered in Theorem 1, by using the identity

$$\cot\pi(x_r-x_s) = \tfrac{1}{2}\left(\exp\left(\pi i(x_r-x_s)\right)+\exp\left(\pi i(x_s-x_r)\right)\right)\csc\pi(x_r-x_s).$$

Now we have only to apply Theorem 1 twice, with u_r replaced by $u_r \exp(\pm \pi i x_r)$, in order to obtain Corollary 1.

To prove Theorem 2 we note that if x is any member of a bounded interval, then $\|\varepsilon x\| = \varepsilon|x|$ whenever ε is sufficiently small. Moreover,

$$(\lambda_r - \lambda_s)^{-1} = \lim_{\varepsilon \to 0} \pi\varepsilon \operatorname{cosec} \pi\varepsilon(\lambda_r - \lambda_s)$$

so that we can appeal to Theorem 1.

We deduce Corollary 2 by squaring out the integrand and integrating the terms separately. The diagonal terms contribute $T \sum|a_r|^2$ and the non-diagonal terms may be written as the difference of two bilinear forms of the sort in Theorem 2. A double appeal to that theorem then produces the desired result.

Our remark after Corollary 2 makes it clear that it suffices to prove Corollary 3 in the case of finite sums. The result then follows from (1.8) on noting that $|\log(m/n)| \geqslant 1/(n+1)$ for $m \neq n$.

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